VEGO GLOBAL Exploratory Data Analysis: Project Report

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This exploratory data analysis considers VEGO GLOBAL's sales, customer, product, and geographical data to identify patterns for sales performance, profitability, customer understanding, and business decision-making. The dataset contains 362,388 non-null sales records covering the period from December 2010 to January 2014 and was constructed by integrating sales transactions with customer, product, and geographical information. Derived indicators, including Profit and Profit Margin, were calculated to evaluate financial performance.

he dataset contains 362,388 sales records, generating approximately 176.15 million $ in sales and 72.49 million$ in profit, with an overall profit margin of 41.15 %. The analysis highlights strong geographical and product differences, with the United States and Australia accounting for about 63% of sales and Road Bikes generating the highest sales. Customer income shows little relationship with order value, while a negative correlation between sales amount and profit margin suggests that higher-value sales tend to have lower percentage margins. Promotional transactions show slightly lower margins than non-promotional ones, while sales display clear seasonal variation. These findings support recommendations focused on regional diversification, higher-margin products, improved inventory planning, and better alignment of promotions and operations with seasonal demand.

Repository Contents

File	Purpose
`sales.py`	Cleans and reshapes the raw Sales export
`Customersheet.py`	Cleans the raw Customers export
`ProductEnglish.py`	Cleans the raw Product Catalog export
`Geography.py`	Cleans the raw Geography export
`json_to_csv.py`	Converts a raw JSON source (customers) to CSV
`merged.py`	Merges the four cleaned tables into one
`EDA.ipynb`	Full exploratory analysis notebook
`Unified_Sales_Data.zip`	Extracted form of the final database

1 Introduction

VEGO GLOBAL's sales database contains information on transactions, customers, products, and geographical areas. Such data can provide useful information about the company's sales performance and customer purchasing patterns. However, the presence of several interconnected tables means that the information must first be organized and examined before meaningful conclusions can be drawn.

Exploratory Data Analysis (EDA) provides a suitable approach for this purpose. It allows the main characteristics of the data to be examined through descriptive statistics and visualizations, while also helping to identify differences between customer groups, products, and geographical areas. In a business context, these observations can support a better understanding of sales performance and help identify patterns that may deserve further investigation.

A large sales database does not directly provide an explanation of the factors associated with sales performance. The main challenge is therefore to transform the available transactional and descriptive data into interpretable information.

Problem Statement & objective

The central question of this analysis is:

> What patterns and relationships can be identified in VEGO GLOBAL's sales data, and what hypotheses about its business performance can be generated from these patterns?

The analysis focuses on describing the data and identifying meaningful patterns rather than establishing causal relationships.

Thus, the project objective is to Analyze VEGO GLOBAL's business data and produce data driven insights to improve sales strategies, optimize marketing and targeting efforts, evaluate regional sales performance, and support financial reporting.

Scope

The analysis is based on the sales, customer, product, and geographical data provided for the project. The work includes data preparation, descriptive analysis, visualization, comparison of groups, and exploration of relationships between variables.

The study is intentionally limited to an exploratory and descriptive perspective, it assumes `df` the cleaned dataset is already loaded in the session. Libraries used: `pandas`, `numpy`, `Matplotlib` and `seaborn` only. The results of the EDA are used to identify the most relevant patterns and formulate hypotheses that can serve as a basis for subsequent statistical analysis or more advanced modeling.

2 Raw Data Pretreatment

The dataset originates from four separate excel/Parquet/CSV/JSON extracts, structured as a *star schema*: one fact table (transactions) and three dimension tables.

Table	Role	Key
Sales details	Fact table — one row per order line	`SalesOrderNumber`
Customers details	Dimension — customer demographics	`CustomerKey`
Product catalog details	Dimension — product attributes & pricing	`ProductKey`
Geography details	Dimension — sales territory attributes	`SalesTerritoryKey`

Cleaning steps applied to each source

- **Sales:** the raw sales export stored `SalesAmount` split across six currency-coded columns (USD, AUD, CAD, DEM, EUR, GBP). These were unpivoted with `pandas.melt()` into a single `CurrencyKey` / `SalesAmount` pair. Missing USD amounts were backfilled from `UnitPrice`. Non-essential columns (`RevisionNumber`, `UnitPriceDiscountPct`, `DiscountAmount`) were dropped, and `OrderDate` / `ShipDate` were cast to proper datetime types.
- **Customers:** localized name/description columns (Spanish/French variants) were dropped, missing `MiddleName` values were labeled explicitly, and `Gender`, `MaritalStatus`, and date fields were cast to appropriate dtypes.
- **Product catalog:** duplicate rows were removed, non-English description columns were dropped, missing `Status` was set to "Current," missing `Size` was standardized, letter sizes (S/M/L/XL) were mapped to numeric values, and missing `Color` / `Class` / `ProductLine` were labeled "Not Specified."
- **Geography:** the redundant `SalesTerritoryAlternateKey` column was removed.

Merge strategy

The four cleaned tables were joined with `pandas.merge()`, using *left joins* anchored on the Sales fact table:

```
Sales -> (left join) Customers   on CustomerKey
      -> (left join) Product     on ProductKey
      -> (left join) Geography   on SalesTerritoryKey
```

All keys matched fully across tables every sales row resolved to exactly one customer, product, and territory record, with zero missing joins. The result is a single unified table of *362,388 rows* $\times$ *53 columns*, saved as `Unified_Sales_Data.csv`.

Derived analysis fields

Field	**Formula**	**Purpose**
`Profit`	`SalesAmount, TotalProductCost`	profitability
`ProfitMargin`	`Profit / SalesAmount * 100`	Percentage profitability
`OrderYear/Month/Quarter`	Extracted from `OrderDate`	Time-based analysis

3 Exploratory Data Analysis

3.1 Overall Business Performance

Measured Cards

Metric	**Value**
Data period	Dec 29, 2010 – Jan 28, 2014
Total Sales	**$176,152,063**
Total Cost	$103,666,761
Total Profit	**$72,485,302**
Overall Profit Margin	**41.1%**
Number of Orders	27,659
Number of Customers	18,484
Number of Products	158

Monthly sales climbed steadily across 2011–2013, with the sharpest jump in 2013 (a large increase in both order count and revenue versus 2011–2012), before the partial 2014 data cuts off. Average profit margin held broadly stable month to month, indicating that growth in sales volume did not come at the cost of eroding margin

3.2 Customers Insight

- **Occupation:** Professionals ($59.4M) and Skilled Manual workers ($38.6M) are VEGO GLOBAL's largest customer segments by sales, and together anchor the core customer base. Margin percentages are fairly flat across occupation groups ($\sim$52 to 54%), so the differentiator is customer volume and spend, not margin.

- **Education:** Bachelors ($59.4M) and Partial College ($46.3M) customers generate the most total sales and together make up over half of the customer base, though Partial High School customers show the single highest average margin (55.3%) a smaller but proportionally profitable segment.
- **Home ownership:** the majority of customers are homeowners, VEGO GLOBAL's largest customer segment by this attribute.
- **Household composition:** most sales come from customers with no children at home ($107.6M of the total); among customers with children, spending does not increase or decrease in a simple straight line with the number of children.
- **Gender:** total sales are nearly identical between male and female customers gender does not appear to be a meaningful driver of purchasing behavior in this dataset.
- **Cars owned:** customers owning 2 cars represent the most valuable segment by sales volume; customers owning 3 to 4 cars contribute the least.

3.3 Product performance

- **Product line:** Road bikes generate the most total sales ($87.7M) but carry a lower average margin (∼49.7%); Mountain bikes generate less total revenue ($61.5M) but a notably **higher average margin (∼57.6%)**. Volume leadership and margin leadership do not coincide.
- **Top products:** the ten best-selling individual products are all Mountain-200 and Road-150 model variants, several exceeding $7 to $8M in sales each a small set of SKUs is responsible for a large share of total revenue.

3.4 Promotion Flag

`PromotionKey = 1` meaning No Promotion applied covers ∼96% of all transactions and is treated as the "No Promotion" baseline; codes 2, 13, and 14 are grouped as "Promotional." The promotional segment is small ($12.3M of $176.2M total sales) and shows a slightly lower average margin (50.8%) than the non-promotional baseline 53.4% promotions do not currently show a clear sales or margin benefit.

3.5 Regional performance

- **Country:** the United States ($56.3M) and Australia ($54.4M) are effectively co-leading markets, together contributing roughly 63% of total sales. The UK, Germany, and France form a solid mid tier ($15.9M–$20.4M each), and Canada trails in total sales ($11.9M) despite posting the highest average margin of any country (56.4%).

- **Region (US sub-regions + countries):** within the US, Southwest ($34.3M) and Northwest ($21.9M) drive the bulk of American sales, though the small Northeast and Central regions post the highest average margins (57.4% and 56.5% respectively) high margin, but on very low volume.

- **Territory group:** North America ($68.2M) is the top-performing group overall, ahead of the Pacific ($54.4M) and Europe ($53.6M) groups, on every measure total sales, total profit, and average margin.

3.6 Relationships Explore

Note: *Correlation does not imply causality these are directional signals for further investigation, not proof of cause and effect.*

Relationship	Correlation (r)	Interpretation
SalesAmount ↔ UnitPrice/Cost	≈ 1.0	Structural — single-unit order lines
ProfitMargin ↔ SalesAmount	≈ -0.47	Higher-priced items carry lower % margin
ProfitMargin ↔ TotalProductCost	≈ -0.48	Margin % shrinks as cost/price rises
Profit ↔ ProfitMargin	≈ -0.44	High-volume, lower-margin items still profit
YearlyIncome ↔ SalesAmount	≈ 0.07	Income is a weak predictor of order value

3.7 Time Analysis

Excluding the partial 2010 and 2014 data, sales grow steadily through the calendar year, accelerate into *Q4 (Oct–Dec)* the strongest quarter overall at $54.6M and show a secondary bump in *June*. This pattern held for both total sales value and number of orders, suggesting genuine demand seasonality rather than a pricing artifact.

General Insights

- VEGO GLOBAL generated approximately $176.2M in total sales and $72.5M in profit over the observed period, corresponding to an overall profit margin of 41.1%.

- The United States and Australia are the leading individual markets, together contributing approximately 63% of total sales.

- North America is the highest-performing sales group, followed by the Pacific region and Europe.

- Road bikes generate the highest total sales, while Mountain bikes show the strongest profit margin performance.

- Promotions account for only 4% of transactions and show no clear margin advantage compared with non-promotional sales.
- Customer income appears to have limited importance when targeting consumers, suggesting that other customer characteristics may be more relevant for segmentation.
- Correlation analysis reveals associations between selected numerical variables, helping to identify variables that may be related to sales performance. However, these relationships should not be interpreted as evidence of causality.
- sales Grow monotonically through the year with an a faster growth during the 4th quarter *Q4 (Oct-Dec)*, and a secondary peak in June, this seasonal pattern can be used as a sales banchmark and should be studies and enphesised.

4 Hypotheses Generated from the EDA

1. **Uneven regional performance / concentration risk.** Revenue is heavily concentrated in the US and Australia, with the UK/Germany/France cluster and Canada trailing well behind. Over-reliance on two markets increases exposure to local economic or competitive shocks. *Next step:* examine growth trends by country over time and evaluate the investment case for underinvested markets.
2. **Higher-value items carry structurally lower percentage margins.** VEGO GLOBAL's pricing/costing structure yields lower percentage margins on higher-priced products, even though absolute profit per unit is higher. *Next step:* review markup formulas by product line/class for room to raise margin on high-ticket items.
3. **Promotions are not currently a meaningful growth or margin lever.** Promotional transactions do not outperform non-promotional ones on margin, and represent a very small share of overall sales ($\sim$4%). *Next step:* run a controlled before/after comparison on a specific promotion.
4. **Customer income is a weak predictor of order value.** Yearly income has almost no relationship with per-order spend ($r \approx 0.07$). *Next step:* explore combined segmentation (occupation + region + product interest).
5. **Sales exhibit clear, recurring seasonal demand.** VEGO GLOBAL shows a repeating Q4 peak with a secondary mid-year (June) bump. *Next step:* build a seasonal inventory and staffing plan aligned to these peaks.
6. **Non-price demographic factors segment customer value more than gender.** Gender shows almost no difference in total sales, while household composition and car ownership

show more separation. *Next step:* build customer segments around lifestyle/household attributes.

5 Recommendations for the Company

1. **Diversify regional exposure.** Invest in growing mid-tier and underperforming markets (Canada, and the small US Southeast/Northeast/Central regions) to reduce reliance on the US/Australia concentration.

2. **Rebalance product marketing toward margin.** Shift a portion of marketing spend from high-volume Road bikes toward higher-margin Mountain bikes.

3. **Protect top-SKU availability.** Tighten inventory planning (safety stock, reorder points) around the top 10 revenue-driving Mountain-200 / Road-150 variants.

4. **Reassess promotion strategy.** Redesign or retarget promotions, since the current promotional segment shows no clear margin or scale benefit.

5. **Align operations to seasonality.** Plan inventory build-up, staffing, and marketing pushes ahead of the Q4 peak and the secondary June uptick.

6. **Segment customers beyond income and gender.** Use occupation, household composition, and car ownership to refine targeting.

6 Limitations and Contribution Horizons

Data Quality and Missing Values

Although the exploratory analysis provides several useful insights into VEGO GLOBAL's business data, some limitations should be considered when interpreting the results.

Missing values were present in several variables of the original dataset. Their treatment depended on the nature and proportion of the missing observations. Variables with a high proportion of missing values were examined carefully before being retained, transformed, or excluded from particular analyses. Consequently, some results are based only on observations for which the required variables were available. The presence of missing data may therefore affect the representativeness of certain comparisons and should be considered when interpreting the results.

Interpretation of Variables and Coded Values

The meaning of each variable was determined from its column name, its data type, its observed values, and its relationship with the other tables. Particular attention was required for variables represented by numerical codes rather than descriptive labels. For example, binary variables coded as 0 and 1 were interpreted according to the corresponding variable definition, where 0 generally represents the absence of a characteristic and 1 its presence. These values were therefore not treated as continuous numerical measurements.

Similarly, variables such as `CustomerKey`, `ProductKey`, and `SalesTerritoryKey` were treated as identifiers rather than quantitative variables. In contrast, variables such as `UnitPrice`, `SalesAmount`, and profit-related measures were interpreted as numerical business indicators. This distinction is important because performing statistical calculations on identifiers or categorical codes could produce misleading results.

Where descriptive labels were not directly available, the interpretation of certain coded values was based on the structure and distribution of the data. Such interpretations should ideally be verified against the original data dictionary or corresponding dimension tables.

The temporal analysis is also affected by incomplete boundary years. The years 2010 and 2014 contain only one month of observations and were therefore excluded from some seasonal and year-over-year comparisons. A more complete historical dataset would provide stronger evidence for identifying long-term trends and seasonal patterns.

Interpretation of Relationships

The relationships identified through correlation and group comparisons should be interpreted as associations rather than causal effects. For example, a relationship between sales and another business variable does not by itself establish that changes in that variable cause changes in sales. The hypotheses generated from the EDA would therefore require further statistical testing or controlled analysis before being used to support major pricing, marketing, or inventory decisions.

Contribution Horizons

Despite these limitations, the present analysis provides a foundation for several possible extensions. First, obtaining more complete data and an explicit data dictionary would improve the interpretation and reliability of the variables. Second, recovering missing information such as order quantity and promotion details would allow more precise analyses of product volume, discounting, and promotional effectiveness.

The results also provide a basis for more advanced statistical analysis. The hypotheses generated from the EDA could be tested using formal statistical methods, while the temporal patterns identified could be developed into sales forecasting models. Customer and product characteristics could additionally be used for segmentation, profitability analysis, or predictive modeling.

Finally, the analysis can contribute to a broader business intelligence process by transforming raw transactional data into measurable indicators, identifying areas requiring further investigation, and providing a starting point for future forecasting and decision-support applications.

7 Conclusion

This study used Exploratory Data Analysis (EDA) to examine VEGO GLOBAL's sales data and identify patterns that can contribute to a better understanding of the company's business performance. By combining sales, customer, product, and geographical information, the analysis provided a descriptive view of the main characteristics of the business and highlighted differences across markets, product categories, and customer groups.

The results show a concentration of sales across a limited number of markets, with the United States and Australia representing an important share of total sales. At the regional level, North America was identified as the strongest sales group. The product analysis also revealed differences between sales performance and profitability, with Road bikes leading in total sales while Mountain bikes showed stronger margin performance. In addition, the analysis of promotional activity did not indicate a clear margin advantage for promotional transactions, while customer income appeared to provide limited explanatory value for consumer targeting. The temporal and correlation analyses further helped identify patterns and associations that may deserve more detailed investigation.

Beyond these individual findings, the main contribution of the EDA is the generation of a set of hypotheses concerning VEGO GLOBAL's products, customers, markets, and commercial practices. These hypotheses provide a starting point for moving from descriptive analysis toward more formal statistical investigation. However, the presence of missing values, coded variables, incomplete periods, and limitations in some available fields means that the findings should be interpreted within the scope of the available data.

Overall, the analysis demonstrates how a structured EDA process can transform a complex sales database into interpretable business information. The results provide a foundation for future work involving statistical hypothesis testing, customer and product segmentation, profitability analysis, and sales forecasting. With more complete data and clearer documentation of variable definitions and coding, these extensions could provide deeper evidence to support VEGO GLOBAL's future business and decision-making processes.

Appendix: Repository Resources & Code

A. Correlation Matrix

```
corr = df[numerical_columns].corr()

plt.figure(figsize=(10, 7))
sns.heatmap(corr, annot=True, cmap="coolwarm")
```

```
plt.title("Correlation Matrix")
plt.show()
```

B. Missing Values Summary

```
missing = df.isnull().sum()
missing = missing[missing > 0].sort_values(ascending=False)
print(missing)
```

C. Group Comparison

```
grouped = df.groupby("Country")[["SalesAmount", "Profit"]].sum()
print(grouped.sort_values("SalesAmount", ascending=False))
```

D. Distribution of a Numerical Variable

```
plt.hist(df["SalesAmount"].dropna(), bins=30)
plt.xlabel("Sales Amount")
plt.ylabel("Frequency")
plt.title("Distribution of Sales Amount")
plt.show()
```

E. Time-Based Sales Analysis

```
df["OrderDate"] = pd.to_datetime(df["OrderDate"])

monthly_sales = (
df.groupby(df["OrderDate"].dt.to_period("M"))["SalesAmount"]
.sum()
)

monthly_sales.plot()
plt.xlabel("Month")
plt.ylabel("Sales Amount")
plt.title("Monthly Sales Evolution")
plt.show()
```

F. Binary Variable Interpretation

```
df["Variable"].value_counts(dropna=False)
```